MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 4703: NETWORK ADMINSTRATION AND INFORMATION SECURITY

END OF SEMESTER EXAMINATION

DATE: 15 JULY, 2016 **TIME ALLOWED**: 8AM-11AM

INSTRUCTIONS:

i. Answer **ALL THREE** questions
ii. Write your answers clearly
iii. Marks for each question are indicated

QUESTION ONE: GENERAL NETWORKS

a) Define Network Security. **[2 marks]**
b) Define the concept of virtual circuit. **[4 marks]**
c) Explain perimeter defense as a network security measure. **[6 marks]**
d) Explain how the TCP protocol would be able to detect errors and data loss, and how it would ensure that the lost data is re-transmitted, whilst transmitting data over a TCP virtual circuit. **[6 marks]**
e) Symmetric-Key and Asymmetric-Key Cryptography complement each other. Explain. **[9 marks]**

QUESTION TWO: ADMINISTERING GLOBAL NETWORKS

a) Telecommunication companies (Telcos) that provide global network services, define the services they offer to customers in a Service Level Agreement (SLA). Explain the purpose of an SLA and give examples of what it might contain. **[7 Marks]**

b) Companies that have offices in several countries around the globe need to create a private corporate network that is able to connect these sites together and transport traffic of different types between them. An increasingly popular way of providing such a network is to use Multiprotocol Label Switching (MPLS) data services. Briefly explain how MPLS works and how it is able to support different traffic types. **[12 marks]**

c) An alternative to MPLS might be to consider using the Internet. What are the main disadvantages of the Internet that mean it would offer a worse solution than MPLS? **[6 marks]**

QUESTION THREE: ROUTING PROTOCOLS

a) Explain main differences between link-state and distance-vector routing protocols. **[5 marks]**
b) Briefly describe the link-state protocol known as OSPF and explain how it copes with routing inside a large and complex autonomous system. **[6 marks]**

c) You have been given the network address 192.168.2.0/24 to subnet and provide the IP addressing for the network shown in the Topology Diagram. The network has the following addressing requirements:

- The LILONGWE LAN 1 will require 10 host IP addresses.
- The LILONGWE LAN 2 will require 10 host IP addresses.
- The MZUZU LAN 1 will require 10 host IP addresses.
- The MZUZU LAN 2 will require 10 host IP addresses.
- The HQ-BT LAN will require 20 host IP addresses.
- The link from HQ to LILONGWE will require an IP address for each end of the link.
- The link from HQ-BT to MZUZU will require an IP address for each end of the link.

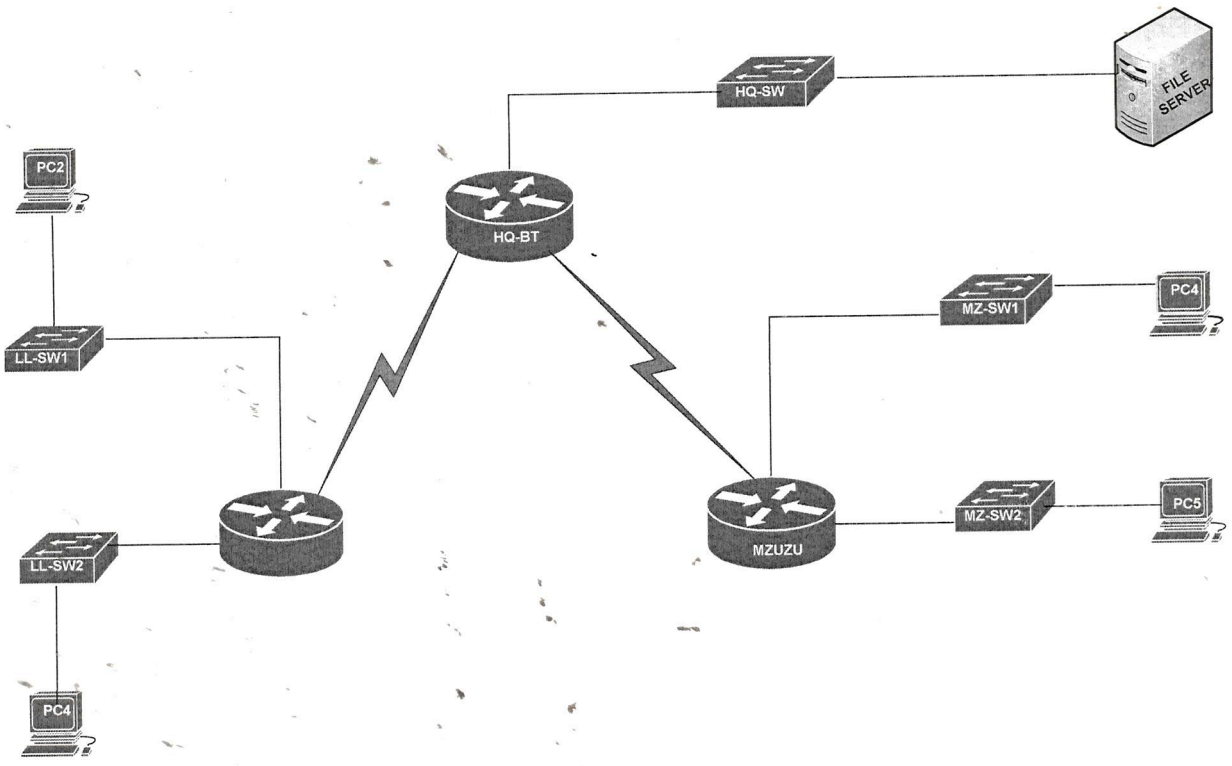


Figure 1: TechNix Malawi

Task 1: Examine the network requirements

Examine the network requirements and answer the questions below. Keep in mind that IP addresses will be needed for each of the LAN interfaces. **[10 Marks]**

i. What will the subnet mask be for the sub networks?

ii. How many subnets are needed?

iii. What is the maximum number of IP addresses that are needed for a single subnet?

iv. How many IP addresses are needed for each of the branch LANs?

v. What is the total number of IP addresses that are needed?

Task 2: Assign IP Addresses to the Network Devices

Assign the appropriate addresses to the device interfaces using appropriate cisco IOS commands. Do not forget to bring up all the interfaces you configure.

Step 1: Assign addresses to the HQ Router **[3 Marks]**

vi. Assign the first valid host address in the HQ LAN subnet to the LAN interface.

vii. Assign the first valid host address in link from HQ to LILONGWE subnet to S0/0/0.

viii. Assign the first valid host address in link from HQ to MZUZU subnet to S0/0/1 interface.

Step 2: Assign addresses to the Lilongwe Router **[3 Marks]**

ix. Assign the first valid host address in the LILONGWE LAN 1 subnet to int. Fa0/0

x. Assign the first valid host address in the LILONGWE LAN 2 subnet to int. Fa0/1

xi. Assign the last valid host address in link from HQ to LILONGWE subnet to the WAN interface.

Step 3: Assign addresses to the MZUZU Router **[3 Marks]**

xii. Assign the first valid host address in the MZUZU LAN 1 subnet to int. Fa0/0

xiii. Assign the first valid host address in the MZUZU LAN 2 subnet to int. Fa0/1

xiv. Assign the last valid host address in link from HQ to MZUZU subnet to the WAN interface.

xv. Implement OSPF routing protocol on the network. **[10 Marks]**

xvi. Assign Fa 0/1 of MZ-SW to Vlan 2 and configure it to access mode, accept one mac-address and shut down if this rule is violated. **[10 Marks]**

END OF QUESTION PAPER